



NETWORK SECURITY GAMES

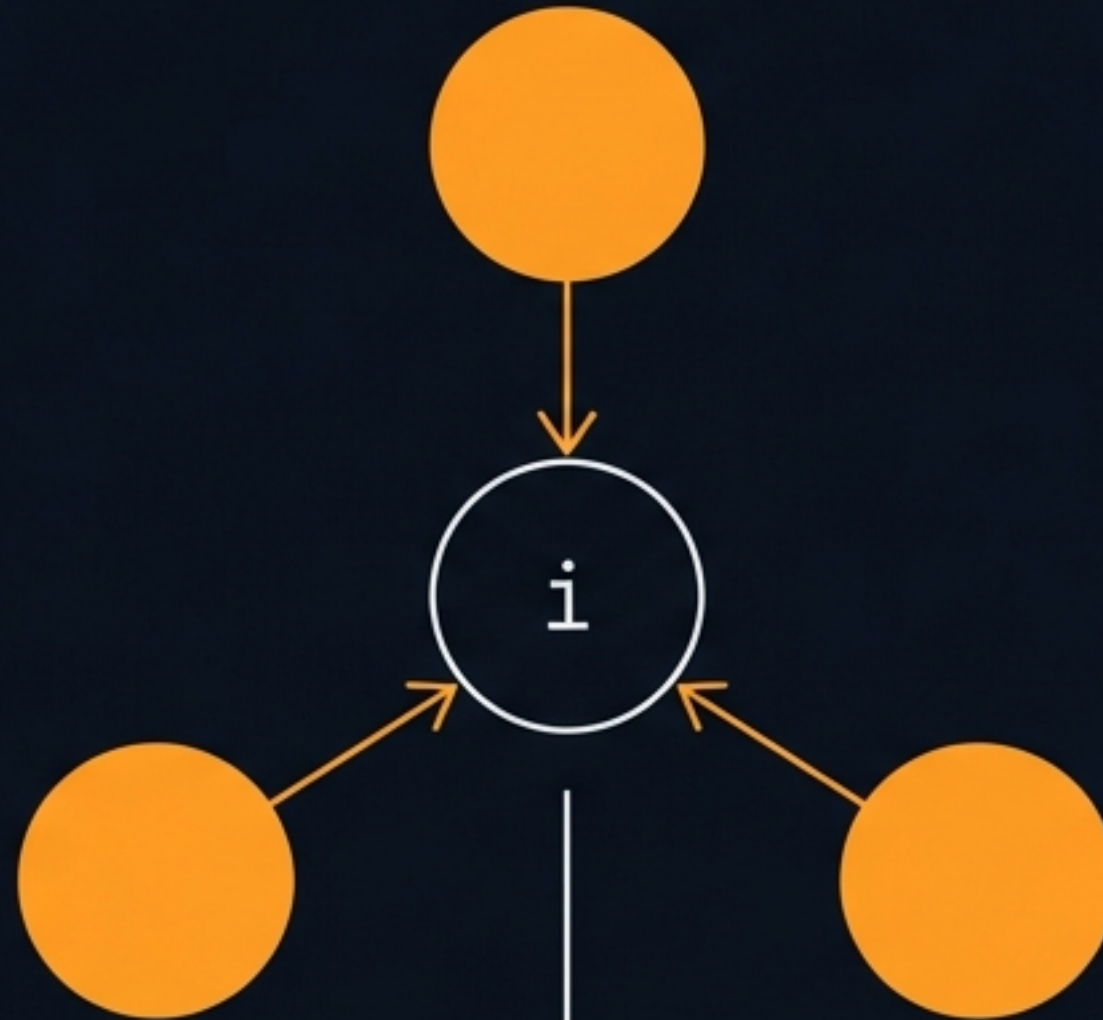
AN ALGORITHMIC GAME THEORY IMPLEMENTATION

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THE PROBLEM: DISTRIBUTED SECURITY

Objective:

Secure graph $G=(V,E)$
covering every edge.



Condition: A node is safe if
covered by ALL neighbors.

The Complexity:

Topologically equivalent to
Minimum Vertex Cover (MVC).

Class: NP-Hard.

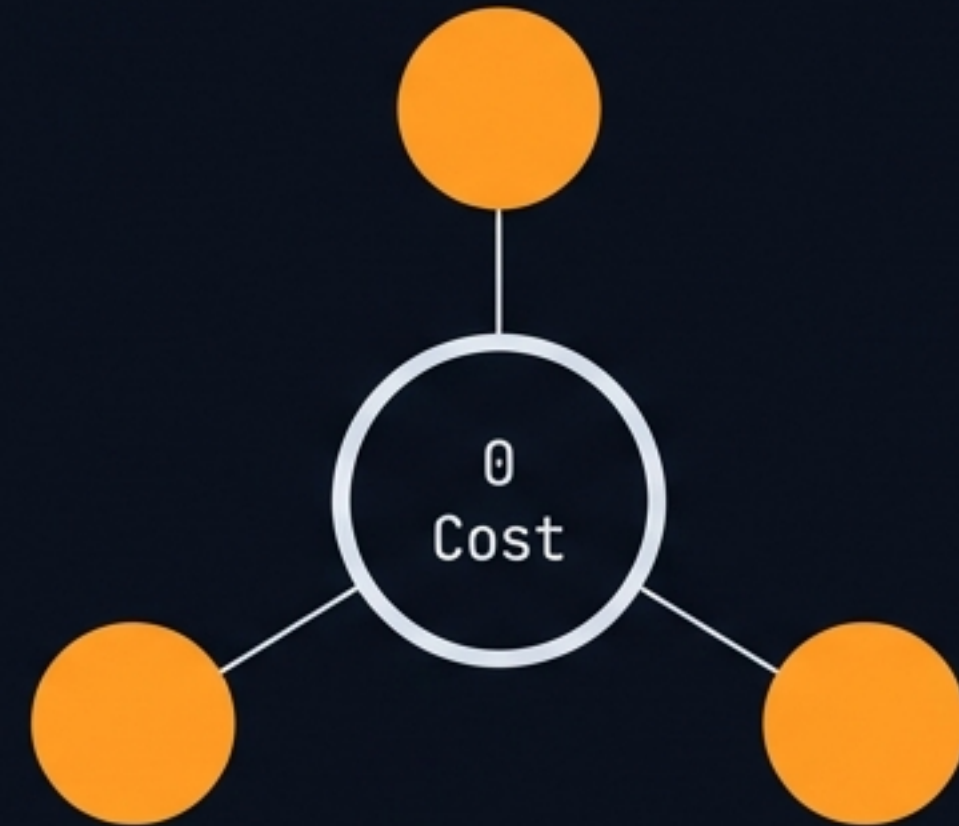
LENS 1: NON-COOPERATIVE DYNAMICS

Strategy: Invest



$$\text{Utility} = \alpha - c \quad (6)$$

Strategy: Free-Ride



$$\text{Utility} = \alpha \quad (10)$$

Insight: Strategic Substitutes. If my neighbor invests, I have less incentive to pay.

LEARNING EQUILIBRIUM

Best Response (BRD)



Myopic Rationality.
Strict improvement
updates only.
 $u_{\text{new}} > u_{\text{old}}$.

Fictitious Play (FP)



History-Based Beliefs
Deterministic Threshold:
Play θ if
 $\text{Probability}(\text{Covered}) >$
 Cost Ratio

Regret Matching (RM)



Minimizing Regret.
Probabilistic selection
based on
counterfactual payoffs.

Result: All methods converge to Pure Nash Equilibria (PNE)

LENS 2: COALITIONAL VALUE

Shapley Value Heatmap



Methodology: Shapley Value (Monte Carlo

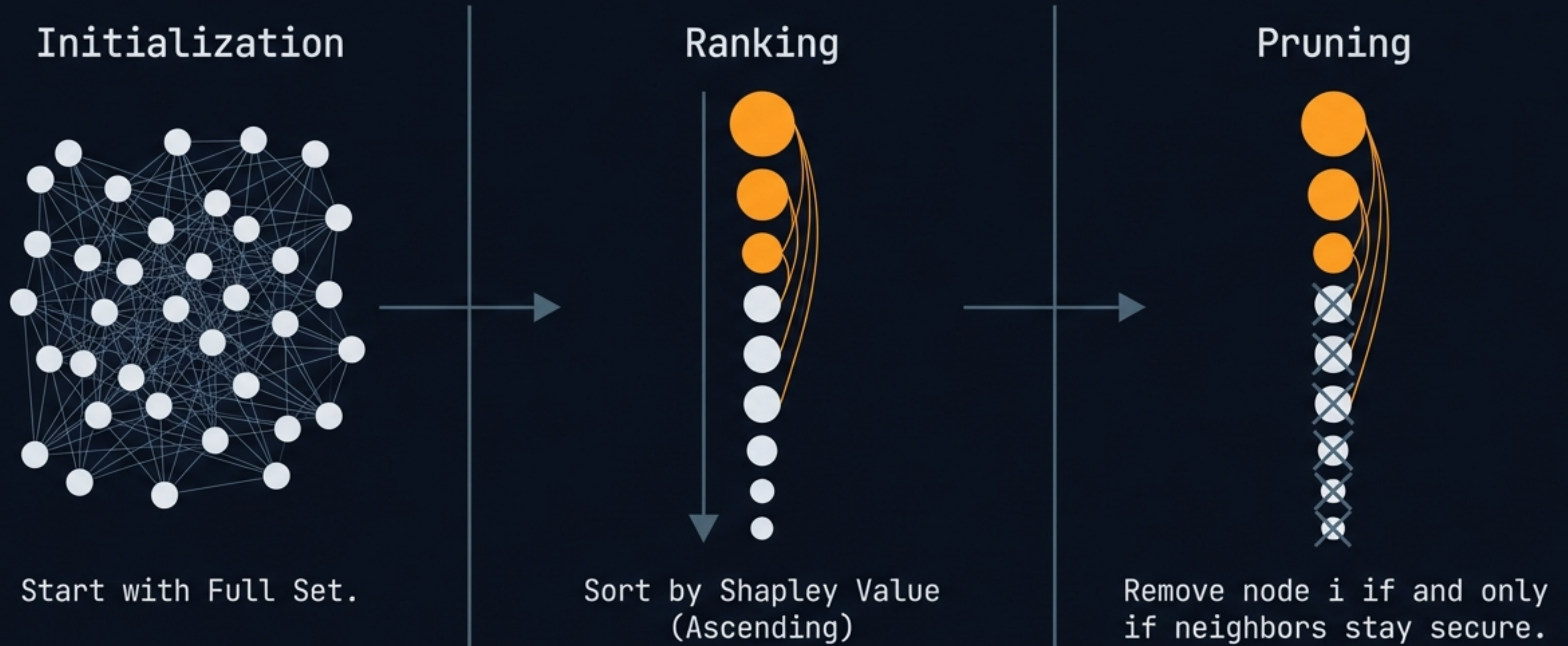
Approximation)

JetBrains Mono Regular (Off-White)

Contribution Logic (+1 Value):

1. Node secures itself upon entry.
2. Node is the LAST missing neighbor to complete coverage for another.

REVERSE GREEDY HEURISTIC



Outcome: High retention of critical hubs. Optimal for scale-free topologies.

LENS 3: THE SECURITY MARKETPLACE

BUYERS (Secure Nodes)



$$\text{Utility} = (\text{Quality} \times 10) + (\text{Budget} - \text{Price})$$

Goal: Maximize Social Welfare

VENDORS (Providers)



MARKET DYNAMICS & SCARCITY

Infinite Capacity

BUYERS



VENDORS
(High Quality)

Pareto Optimal

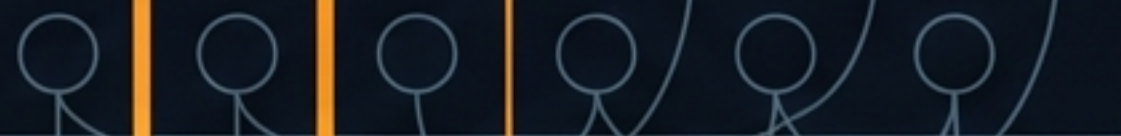
Limited Capacity

BUYERS

HIGH-BUDGET
BUYERS



LOW-BUDGET
BUYERS



BEST
VENDORS



POOR
VENDORS

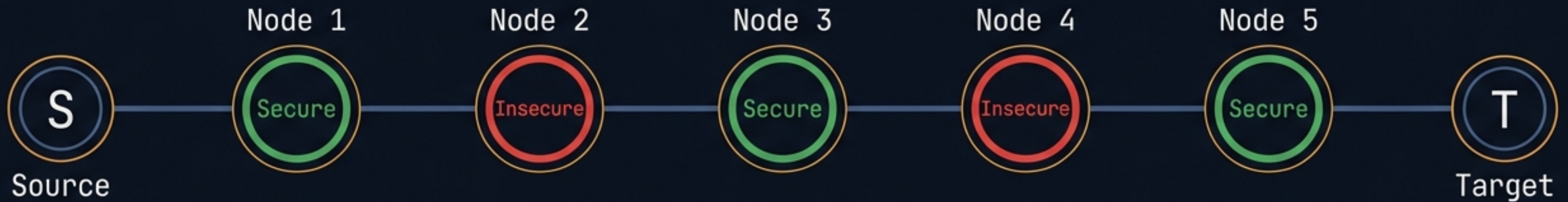


VENDORS

Crowding Out Effect

Wealthy buyers capture quality.
Poor buyers are forced out.

LENS 4: TRUTHFUL PATH AUCTIONS

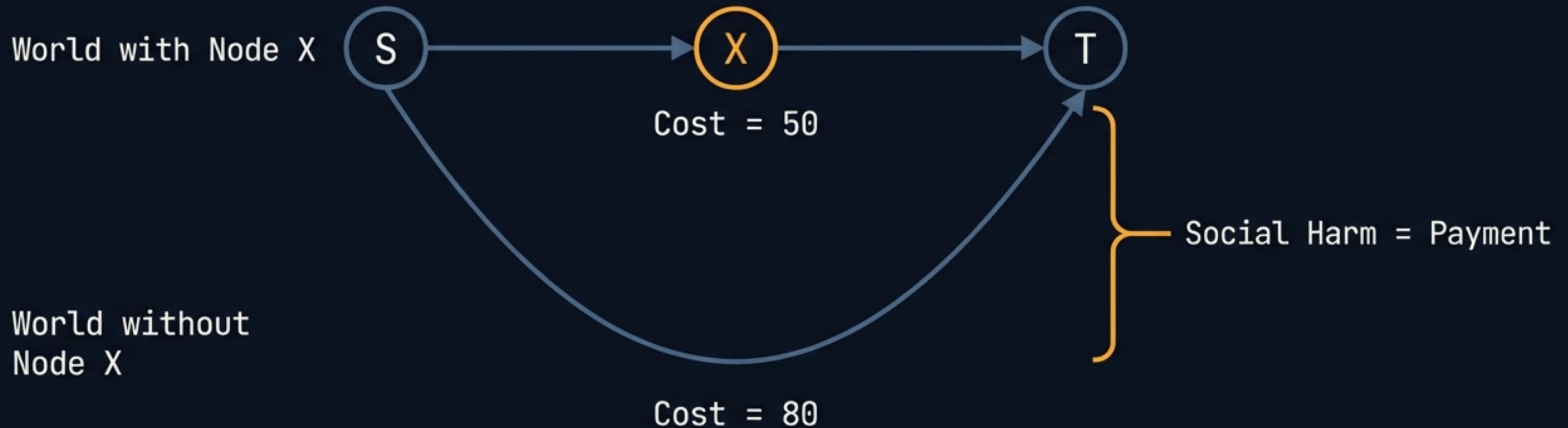


COST FUNCTION

Node Cost (w_i) =	Bid	+	Penalty
			Penalty = 10 (if insecure)
			Penalty = 0 (if secure)

Mechanism: VCG (Vickrey-Clarke-Groves). Truthfulness is the dominant strategy.

PRICING EXTERNALITY (VCG)



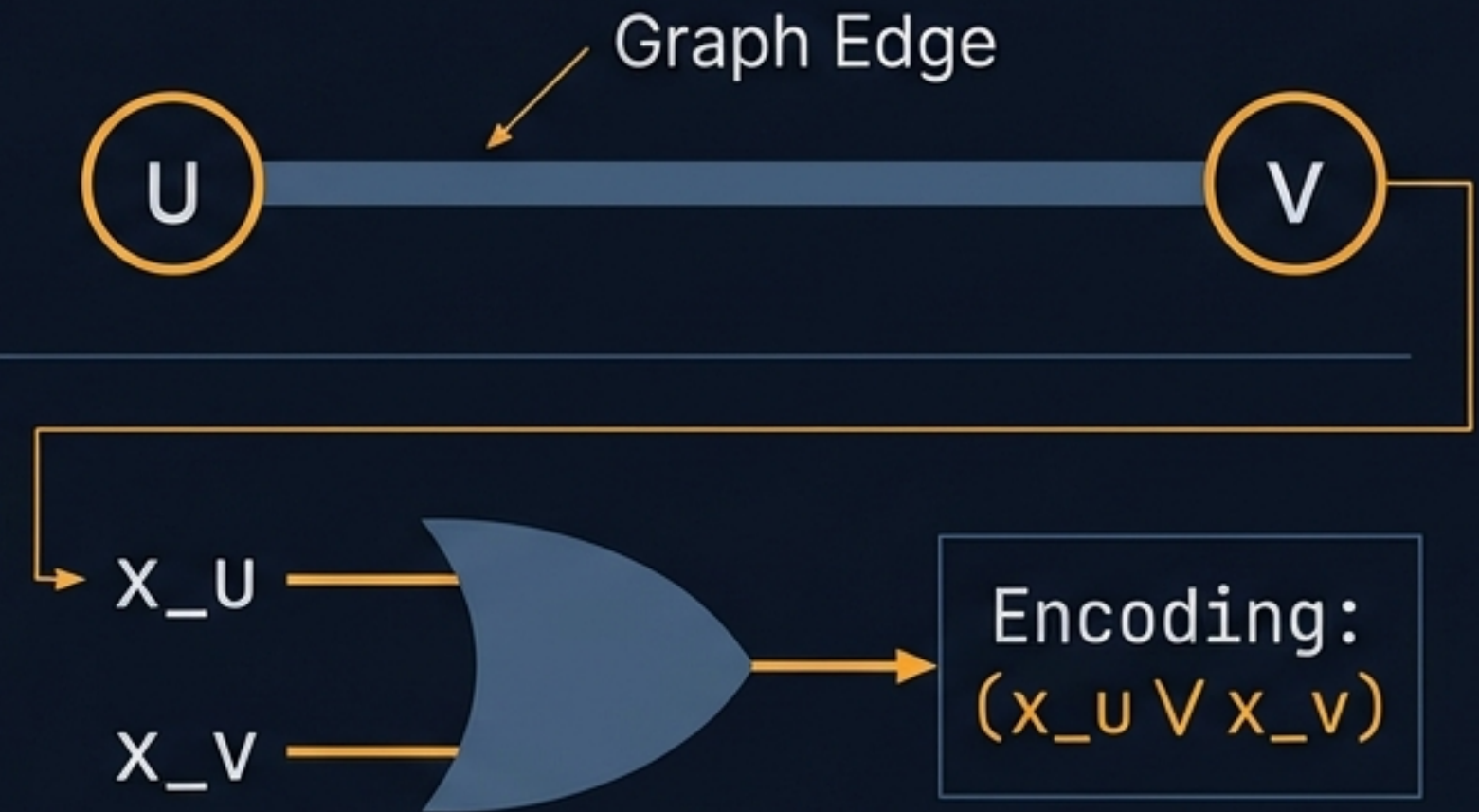
$$\text{Payment}_i = \text{SP}(G \setminus \{i\}) - (\text{SP}(G) - w_i)$$

Note: If removing 'i' disconnects the graph, Payment = Infinity (Monopoly Power).

BONUS: SYMBOLIC SAT SOLVER

Engineering a custom combinatorial solver.

```
formula = And(  
    Implies(x_u, True),  
    Or(x_u, x_v)  
)
```



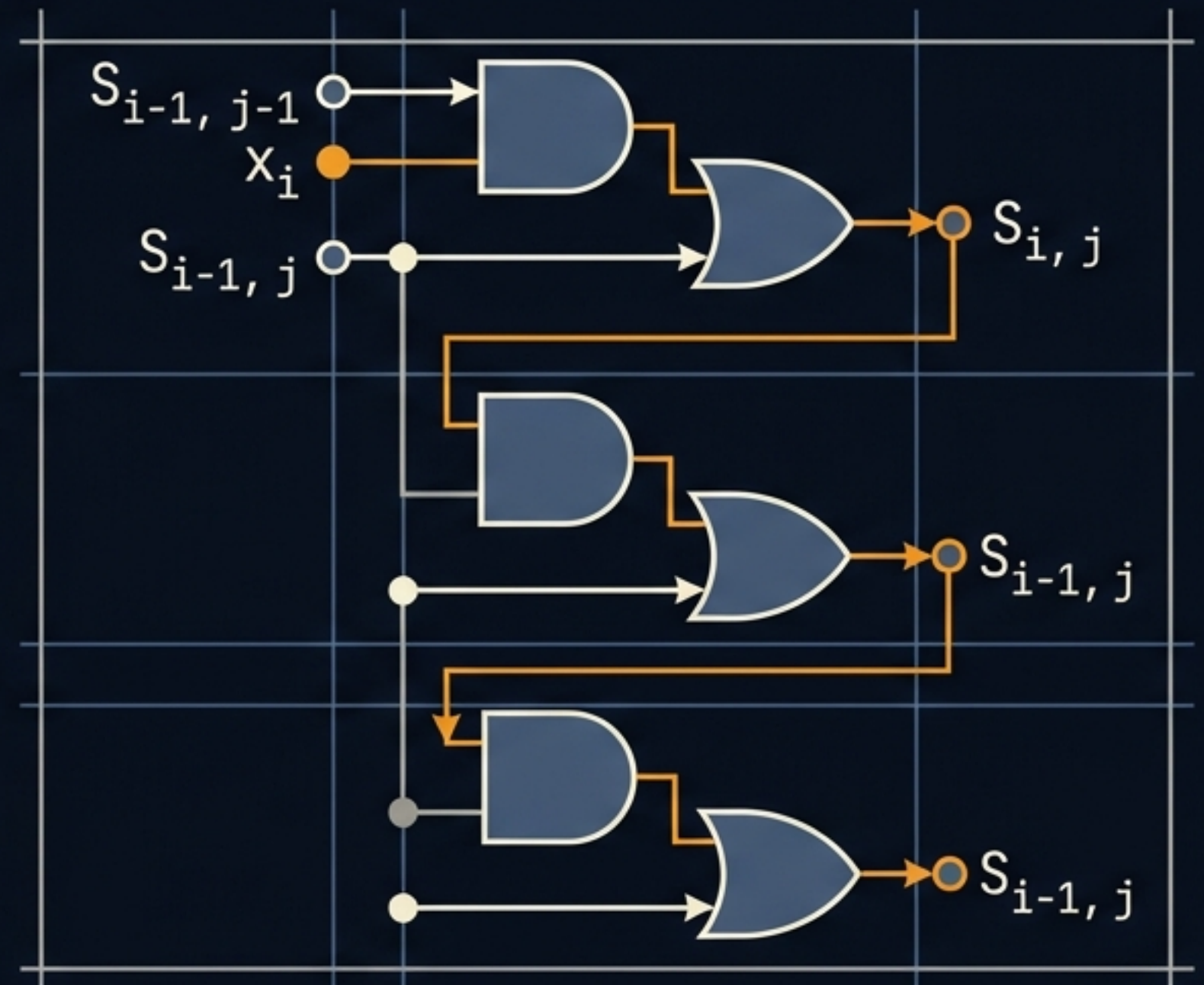
Pivot from Game Theory to Theoretical Computer Science. Built from scratch using SymPy.

THE SEQUENTIAL COUNTER

The Challenge: Boolean logic lacks arithmetic.

The Solution: Recursive circuit to enforce Cardinality ($\sum x_i \leq k$).

$$S_{i,j} \Leftrightarrow S_{i-1,j} \vee (S_{i-1,j-1} \wedge x_i)$$



TOPOLOGY DICTATES STRATEGY

REGULAR GRAPH



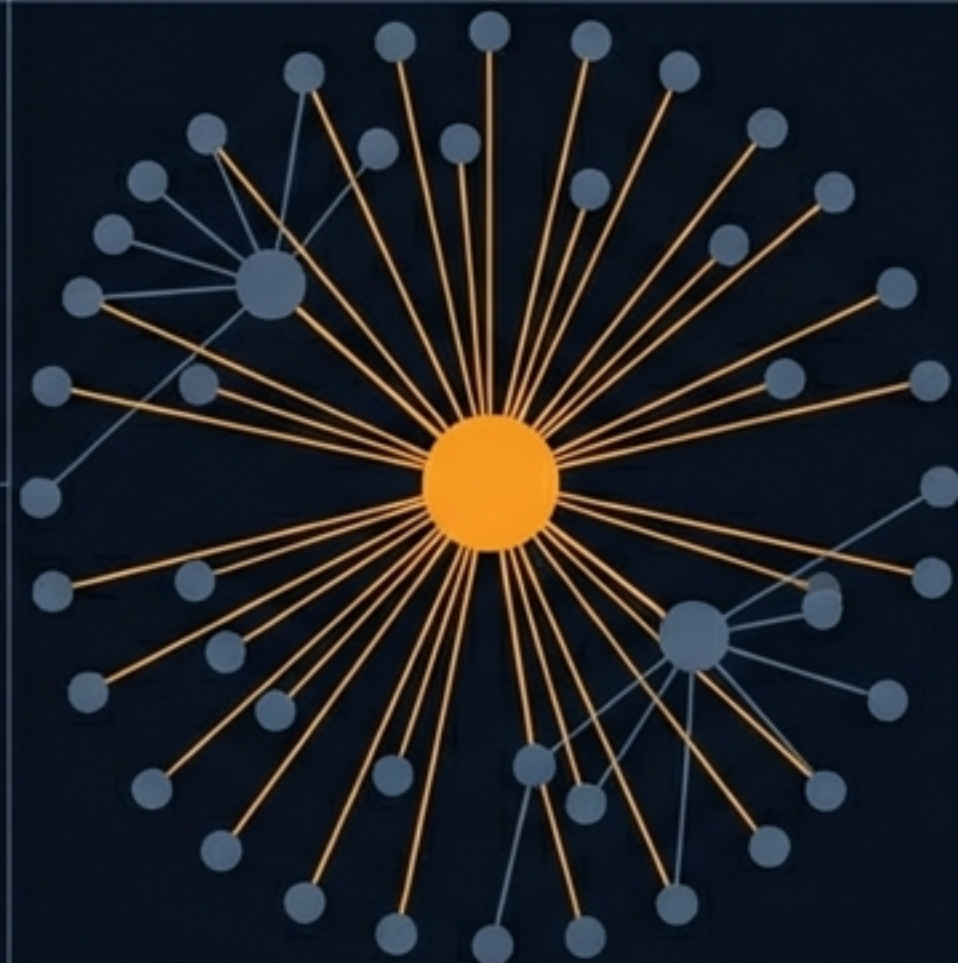
Homogeneous. No leaders.
Costs distributed evenly.

ERDŐS-RÉNYI



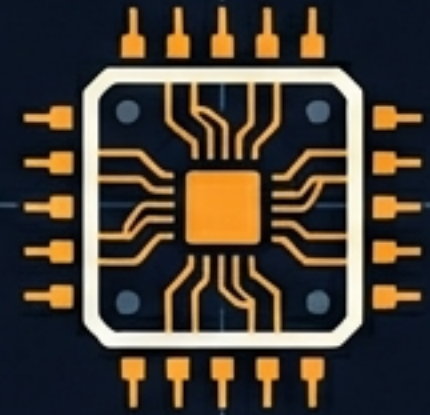
Random Variance. Local
clusters emerge.

BARABASI-ALBERT (SCALE-FREE)



Strict Hierarchy. Hubs are
'Public Good' providers.
Hubs are Monopoly Bridges.

SUMMARY & IMPACT



- Modeled security via Strategic, Cooperative, and Economic lenses.
- Solved equilibrium using Iterative Learning (BRD, FP, RM).
- Designed Truthful Auctions with Externality Pricing.
- Engineered standalone Symbolic SAT Solver using recursive logic.

CONCLUSION: Network topology is the primary determinant of security cost and market power.